



Girijananda Chowdhury University Guwahati 781017, Assam

UNIT TEST - I

BCE23405T Structural Analysis

Session: 2025-2026

Batch: 2024-2028

Date :: 24 February 2026

Total Marks: 20

Time : 1 hour

Question 1: Fill in the Blanks (5 marks)

[CO1]

- In the method of joints, the equilibrium equations used are _____ and _____.
- If the degree of static indeterminacy of a structure is zero, the structure is called _____.
- The strain energy method is based on the principle of _____.
- Castigliano's theorem is applicable to structures behaving within the _____ range.
- The unit load method is commonly used to determine _____ at a particular point in a structure.

Question 2 (5 marks)

[CO1]

- Classify the truss in the figure A below as stable, unstable, statically determinate, or static indeterminate.
- Determine the degree of indeterminacy and the number of degrees-of-freedom for the rigid-jointed frames indicated in figure B.

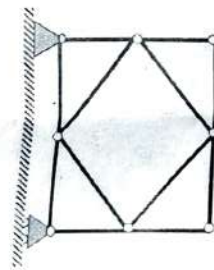


Figure A

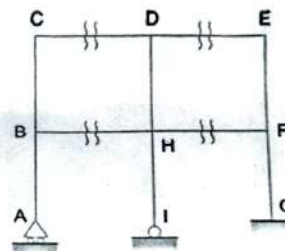


Figure B

Question 3 (10 marks) (Solve any one)

[CO1]

- Use **Unit Load Method** to find the deflection at joint B of the beam shown in Figure 1. Take EI as $50 \times 10^3 \text{ kNm}^2$.
- Determine the vertical deflection at Joint B of the truss shown in Figure 2 using **Castigliano's Theorem**. Take $E = 200 \text{ GPa}$ and $A = 1200 \text{ mm}^2$.

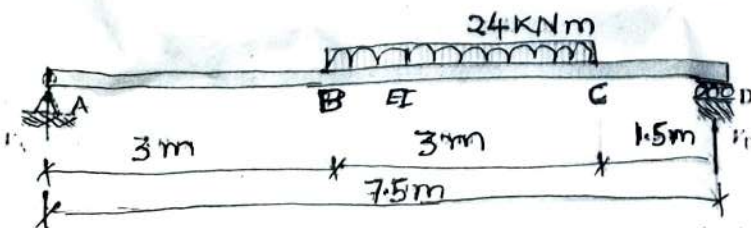


Figure 1

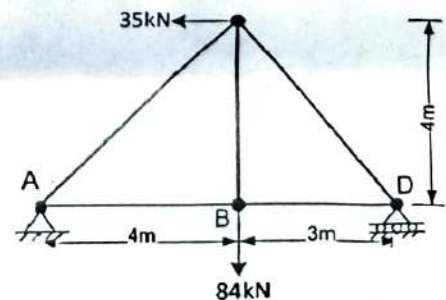


Figure 2



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Girijananda Chowdhury University Guwahati 781017, Assam

UNIT TEST - I (Remedial)

BCE23405T Structural Analysis

Session: 2025-2026

Batch: 2024-2028

Date :: 28 March 2026

Time : 1 hour

Total Marks: 20

Question 1: Fill in the Blanks (5 marks)

[CO1]

- The method of joints is mainly used to determine _____
- When applying the method of sections, the section should cut through Maximum _____ unknown members
- Static indeterminacy of a plane truss is given by _____
- Strain energy stored in a member due to axial load is given by _____
- The partial derivative of strain energy with respect to load gives _____.

Question 2 (5 marks)

[CO1]

- Classify the truss in the figure A below as stable, unstable, statically determinate, or static indeterminate.
- Determine the degree of indeterminacy and the number of degrees-of-freedom for the rigid-jointed frames indicated in figure B.

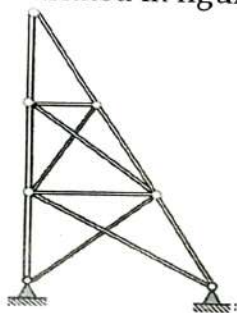


Figure A

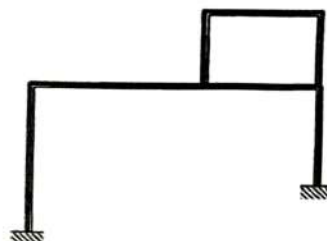


Figure B

Question 3 (10 marks) (Solve any one)

[CO1]

- Consider a simply supported beam ABCD (Figure 1). By taking E is 200 GPa and I is $500 \times 10^6 \text{ mm}^4$. (i) Show that the beam is statically determinate beam. (ii) Using Unit Load Method determine vertical displacement at joint D.
- Determine the vertical deflection at Joint C using Castigliano's Theorem (Figure 2). Take $E = 200 \text{ GPa}$ and $I = 600 \times 10^6 \text{ mm}^4$.

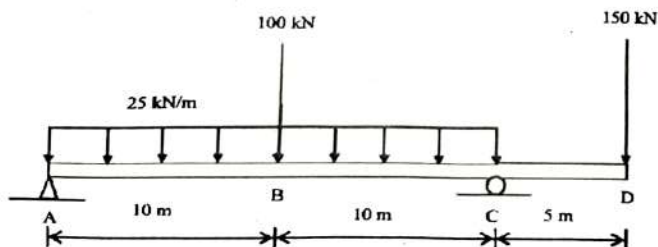


Figure 1

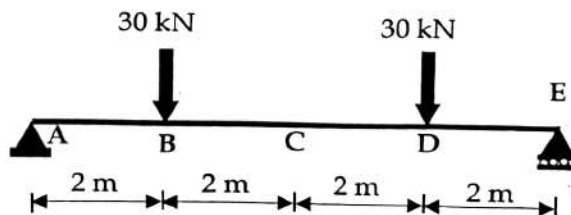


Figure 2